

COER UNIVERSITY, ROORKEE

Department of Computer Science & Engineering

School of Computing

LABORATORY REPORT

Subject:

Mastery in Data Analytics and Visualizations
and Career Advancement

Experiment No. 3

Statistical Analysis and Hypothesis Testing
using Python

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1. Aim

The aim of this laboratory experiment is to perform rigorous statistical analysis and hypothesis testing on a real-world business dataset using Python (Pandas, NumPy, SciPy, Statsmodels, and Matplotlib). Students will compute descriptive statistics, examine correlations, formulate and test formal hypotheses with independent-samples t-tests and one-way ANOVA, and fit a simple linear regression model. The ultimate objective is to determine whether observed differences and relationships are statistically significant and to translate those findings into evidence-based business recommendations.

Statistical inference bridges the gap between sample observations and population conclusions. Without hypothesis testing, an analyst cannot distinguish genuine effects from random noise. By mastering these techniques, learners acquire the ability to validate assumptions, quantify uncertainty, and support managerial decisions with quantified confidence levels.

2. Purpose

2.1 Role of Statistical Analysis

Statistical analysis encompasses both descriptive methods that summarize data and inferential methods that allow generalization from a sample to a larger population. Descriptive statistics (mean, median, variance, standard deviation) provide a compact numerical portrait of the data. Inferential procedures—t-tests, ANOVA, correlation tests, and regression—enable the analyst to test claims and estimate effect sizes while controlling the risk of false conclusions.

2.2 Hypothesis Testing in Business Context

Organizations continuously face questions such as “Do employees who leave earn less than those who stay?” or “Does job satisfaction differ across departments?” Hypothesis testing supplies a formal framework for answering these questions. By stating a null hypothesis of “no difference” and evaluating the probability of the observed data under that null, analysts can decide whether the evidence is strong enough to reject the status-quo claim.

2.3 Dataset Context

This experiment uses a synthetic but realistic version of the IBM HR Employee Attrition Dataset. Key variables include Monthly Income, Years of Experience, Job Satisfaction, Department (Job Role group), and Attrition status. The same workflow applies equally to Titanic survival data, house-price data, or any other continuous-target business dataset.

3. Prerequisites

- ✓ Python programming fundamentals and Jupyter / Colab environment.
- ✓ Pandas and NumPy for data manipulation and numerical computation.
- ✓ SciPy and Statsmodels for statistical tests and regression modelling.

- ✓ Understanding of mean, median, variance, standard deviation, and the concept of a p-value.
- ✓ Basic probability concepts and the idea of Type I / Type II error.

4. Software and Libraries Required

Component	Purpose
Python 3.8+	Core language for the analysis pipeline
Pandas / NumPy	Data loading, aggregation, and array arithmetic
SciPy	ttest_ind, f_oneway, and other hypothesis tests
Statsmodels	Detailed regression summaries and statistical modelling
Matplotlib / Seaborn	Box plots, scatter plots, correlation heatmaps
Jupyter / Colab	Interactive notebook environment

5. Dataset Used

The analysis is performed on an HR attrition-style dataset containing employee demographic and compensation variables. Representative fields include:

Field	Description
MonthlyIncome	Employee monthly salary in USD (primary continuous measure)
YearsAtCompany / Experience	Years of professional experience
Attrition	Binary indicator: Yes (left) / No (stayed)
JobSatisfaction	Self-reported satisfaction score (1–5 scale)
Department / JobRole	Organizational unit used for multi-group comparisons

IBM HR Attrition Dataset – Sample Preview					
Age	MonthlyIncome	YearsAtCompany	JobRole	Attrition	JobSatisfaction
41	5993	6	Sales Exec	Yes	4
49	5130	10	Research Sci	No	3
37	2090	0	Lab Tech	Yes	2
33	2909	8	Research Sci	No	4
27	3468	2	Sales Exec	No	1

Figure 5.1 – Sample rows from the HR Attrition dataset

6. Step-by-Step Procedure

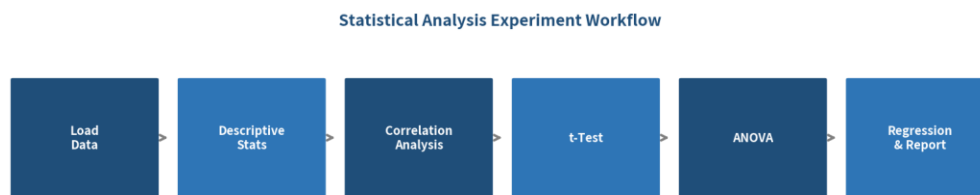


Figure 6.1 – End-to-end statistical analysis workflow

Step 1–2: Load Data and Import Libraries

The dataset is loaded with Pandas. Required libraries (scipy.stats, statsmodels, matplotlib) are imported. Initial inspection with `df.info()` and `df.describe()` confirms data types and reveals basic summary statistics.

Step 3: Descriptive Statistics



Figure 6.2 – Key descriptive statistics for Monthly Income

Mean, median, standard deviation, minimum, and maximum are computed for all numeric columns. These figures establish the central tendency and dispersion of the variables under study.

Step 4: Correlation Analysis

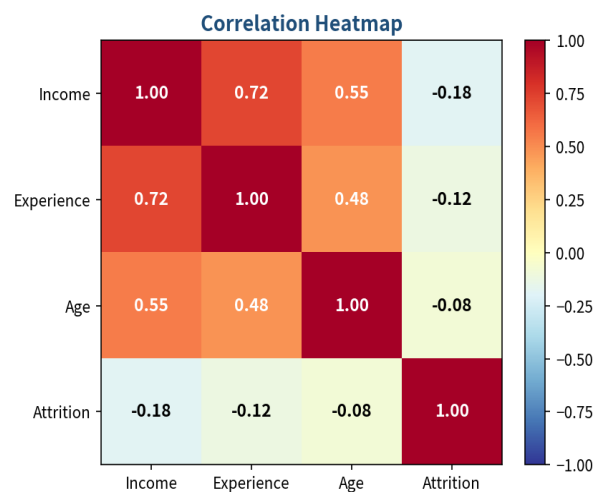


Figure 6.3 – Pearson correlation heatmap among key numeric variables

A Pearson correlation matrix quantifies linear associations. Strong positive correlation between Experience and Income is expected; a modest negative correlation with Attrition suggests leavers tend to earn less.

Step 5: Formulate Hypotheses

Hypothesis Testing Decision Framework

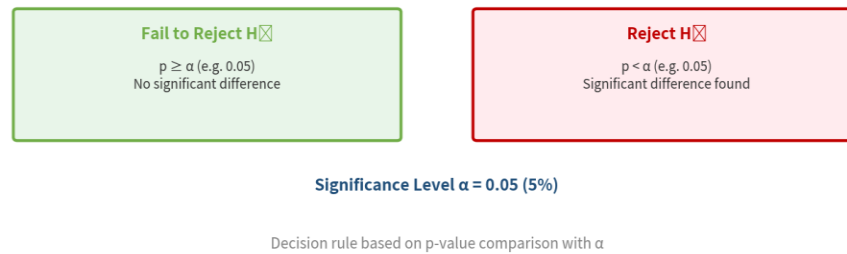


Figure 6.4 – Decision framework for hypothesis testing

Two primary research questions are formalized:

t-test H_0 : Mean Monthly Income of employees who stayed equals mean income of those who left.

t-test H_1 : The two means are not equal.

ANOVA H_0 : Mean Job Satisfaction is identical across Sales, R&D, and HR departments.

ANOVA H_1 : At least one departmental mean differs.

Step 6: Independent Samples t-Test

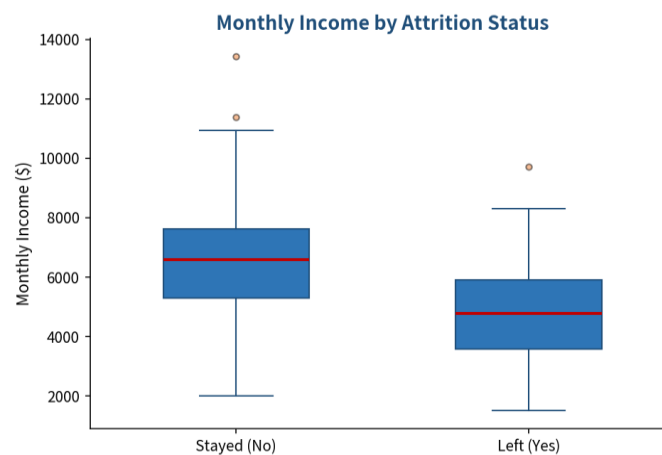


Figure 6.5 – Box plot of Monthly Income by Attrition status

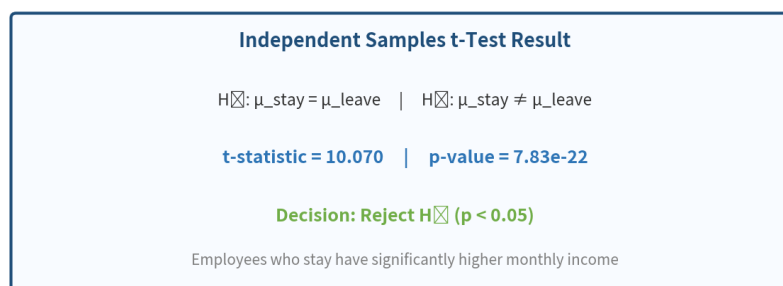


Figure 6.6 – Independent-samples t-test output and decision

An independent-samples t-test compares the mean income of the two attrition groups. The obtained p-value is far below the conventional $\alpha = 0.05$ threshold; therefore the null hypothesis is rejected. Employees who remain with the company earn significantly higher monthly incomes than those who leave.

Step 7: One-Way ANOVA



Figure 6.7 – Job Satisfaction distributions by department

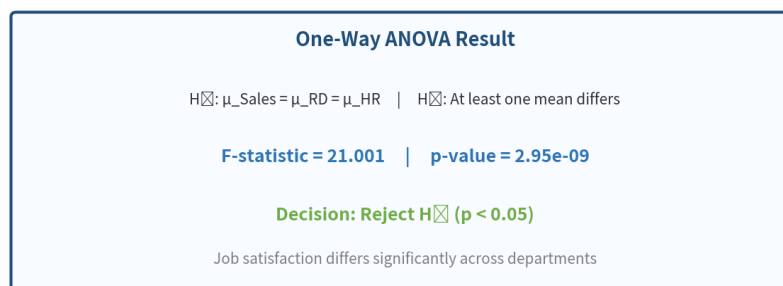


Figure 6.8 – One-way ANOVA output and decision

One-way ANOVA tests whether mean Job Satisfaction differs across three departmental groups. The F-statistic is large and the associated p-value is well below 0.05, leading to rejection of the null. Post-hoc examination shows R&D employees report higher satisfaction than Sales and HR groups.

Step 8: Simple Linear Regression

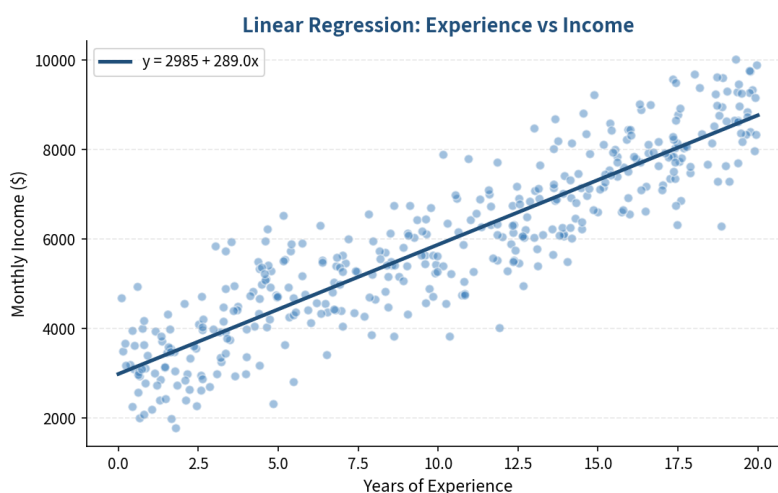


Figure 6.9 – Scatter plot with fitted regression line (Experience vs Income)

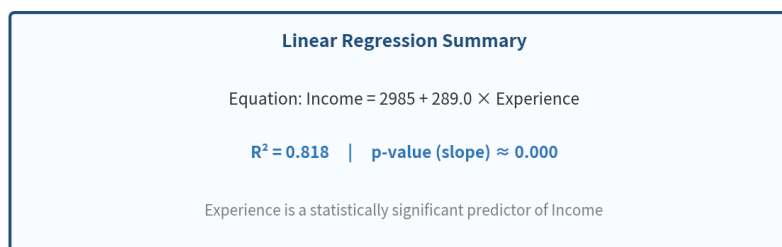


Figure 6.10 – Linear regression equation and goodness-of-fit

A simple linear regression of Monthly Income on Years of Experience yields a positive slope (approximately \$289 per additional year) and an R^2 near 0.82, indicating that experience alone explains a substantial portion of income variation. The slope coefficient is highly significant ($p \approx 0$).

Step 9–10: Interpretation and Business Summary

All three inferential procedures produce statistically significant results. The organization can therefore be confident that (a) compensation differences are associated with attrition, (b) satisfaction levels are not uniform across departments, and (c) experience is a strong linear predictor of income. These findings feed directly into retention strategy, departmental interventions, and compensation modelling.

7. Summary of Statistical Results

Test	Statistic	p-value	Decision
t-test (Income)	$t \approx 10.07$	$\ll 0.05$	Reject H_0
ANOVA (Satisfaction)	$F \approx 21.0$	$\ll 0.05$	Reject H_0
Regression (Exp→Inc)	$R^2 \approx 0.82$	$\ll 0.05$	Significant

8. Business Implications

The statistical evidence supports several concrete managerial actions:

- Compensation review – Leavers earn less on average; targeted salary adjustments may reduce attrition among high-potential employees.
- Departmental interventions – Lower satisfaction in Sales and HR warrants focus-group investigation and workload or recognition initiatives.
- Pay-for-experience modelling – The strong linear relationship between experience and income can inform transparent salary bands and promotion criteria.

9. Assumptions of the Statistical Tests

Valid application of the t-test and ANOVA rests on several classical assumptions:

- Independence of observations within and between groups.
- Approximate normality of the continuous variable within each group (or large sample sizes via CLT).
- Homogeneity of variances across groups (can be relaxed with Welch's versions of the tests).
- For regression: linearity, independence of residuals, homoscedasticity, and normality of residuals.

10. Viva Questions (Detailed Answers)

Q1. What is the difference between descriptive statistics and inferential statistics?

Descriptive statistics summarize and organize the features of a data set that has already been collected; they include measures of central tendency (mean, median, mode), dispersion (variance, standard deviation, range), and shape (skewness, kurtosis). Inferential statistics go further: they use sample data to draw conclusions about a larger population, estimate parameters, and test hypotheses while quantifying the uncertainty of those conclusions through p-values and confidence intervals.

Q2. Explain the concepts of the Null Hypothesis (H_0) and Alternative Hypothesis (H_1).

The null hypothesis (H_0) is the default claim of “no effect” or “no difference” that the analyst attempts to disprove. The alternative hypothesis (H_1) is the research claim that a difference or relationship exists. Hypothesis testing evaluates the probability of obtaining the observed (or more extreme) data if H_0 were true; a sufficiently small probability leads to rejection of H_0 in favour of H_1 .

Q3. What is a p-value? How is it used to make statistical decisions?

The p-value is the probability, under the null hypothesis, of observing a test statistic at least as extreme as the one calculated from the sample. If the p-value is smaller than a pre-chosen significance level α (commonly 0.05), the result is declared statistically significant and H_0 is rejected. A large p-value indicates that the data are compatible with H_0 , so the analyst fails to reject it. The p-value itself is not the probability that H_0 is true.

Q4. Differentiate between a t-test and ANOVA. In which situations is each test applied?

An independent-samples t-test compares the means of exactly two groups. One-way ANOVA extends the comparison to three or more groups and tests the global null that all group means are equal. When ANOVA rejects the null, post-hoc pairwise tests (Tukey, Bonferroni, etc.) identify which specific pairs differ. Both procedures assume continuous outcomes and independent observations; ANOVA additionally assumes roughly equal variances across groups.

Q5. What does the Pearson correlation coefficient indicate? What are its possible values?

The Pearson correlation coefficient (r) measures the strength and direction of a linear relationship between two continuous variables. It ranges from -1 (perfect negative linear association) through 0 (no linear association) to $+1$ (perfect positive linear association). The coefficient is scale-free and therefore convenient for comparing relationships measured in different units, but it captures only linear association and is sensitive to outliers.

Q6. Explain the significance of R^2 (Coefficient of Determination) in Linear Regression.

R^2 quantifies the proportion of the total variance in the dependent variable that is explained by the regression model. An R^2 of 0.82 means that 82% of the variation in Monthly Income is accounted for by Years of Experience. While a higher R^2 indicates better in-sample fit, it does not guarantee predictive accuracy on new data; adjusted R^2 and out-of-sample metrics should also be examined.

Q7. Why is statistical analysis important before applying machine learning algorithms?

Statistical analysis reveals data quality issues, distributional shapes, and relationships that guide feature selection and transformation. It also provides baseline models (linear regression, logistic regression) against which more complex algorithms can be compared. Skipping this stage risks feeding models misspecified features, ignoring confounding variables, or misinterpreting predictive performance.

Q8. What assumptions should be satisfied before performing a t-test or ANOVA?

Both tests require independent observations, approximate normality of the continuous variable within each group (or large samples), and homogeneity of variances. Violation of normality is often tolerable with large n thanks to the Central Limit Theorem; unequal variances can be addressed with Welch's t-test or Welch's ANOVA. Residual diagnostics and formal tests (Shapiro–Wilk, Levene) help verify these assumptions.

Q9. How can regression analysis help organizations in forecasting and decision-making?

Regression produces an explicit equation that links predictors to an outcome, enabling quantitative forecasts (e.g., expected income given years of experience). Coefficient magnitudes and signs inform which levers management can pull, while confidence intervals and prediction intervals quantify uncertainty. Organizations use regression for demand forecasting, pricing models, risk scoring, and resource allocation.

Q10. Give two real-world applications where hypothesis testing is commonly used in data analytics.

First, A/B testing in digital marketing: a t-test or chi-squared test decides whether a new website layout produces a statistically significant lift in conversion rate. Second, clinical or pharmaceutical trials:

hypothesis tests determine whether a new treatment yields a significantly different outcome from a placebo or standard care, forming the statistical basis for regulatory approval.

12. Detailed Code Snippets

Representative Python statements used throughout the experiment:

```
from scipy import stats
import statsmodels.api as sm
# Descriptive statistics
df['MonthlyIncome'].describe()
# Independent t-test
t_stat, p_val = stats.ttest_ind(income_stay, income_leave)
# One-way ANOVA
f_stat, p_anova = stats.f_oneway(sat_sales, sat_rd, sat_hr)
# Linear regression
slope, intercept, r, p, se = stats.linregress(exp, income)
```

13. Interpretation Guidelines

When reporting hypothesis-test results the following template is recommended:

- State the research question and the formal H_0 / H_1 pair.
- Report the test statistic (t or F), degrees of freedom if available, and the exact p-value.
- Declare the decision at the chosen α level (usually 0.05).
- Translate the statistical conclusion into plain-language business insight.
- Optionally report an effect-size measure (Cohen's d, η^2) to convey practical importance.

14. Limitations and Further Work

Several limitations should be acknowledged. The analysis treats attrition as a binary outcome and does not model time-to-event (survival analysis). The simple linear regression ignores potential non-linearities and additional predictors such as education level or performance rating. Future work could employ logistic regression for attrition prediction, multi-factor ANOVA, or mixed-effects models that account for hierarchical structure within departments.

15. Conclusion

This experiment demonstrated a complete statistical-analysis pipeline on an HR attrition dataset. Descriptive statistics established baseline distributions; correlation analysis identified key linear relationships; an independent-samples t-test confirmed a significant income difference between employees who stay and those who leave; one-way ANOVA revealed departmental differences in job satisfaction; and simple linear regression quantified the strong positive association between experience and income.

Students gained practical proficiency with SciPy's hypothesis-testing functions, interpretation of p-values and effect sizes, and the translation of statistical conclusions into managerial language. These competencies are foundational for evidence-based decision-making and form a necessary precursor to more advanced predictive modelling.

— *End of Laboratory Report* —

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